

# Bud seed cues comparison model outlines

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## 1 Methods

1. We used different data and models for budburst and seed germination.
  - (a) We have complete (known chilling, forcing and photoperiod levels) data for 191 angiosperm species for the budburst model, this data following the code used in MoralesCastilla *et al.* 2024 publication, including removal of outlier species and correcting *Acer pseudolatauns* for *Acer pseudoplatanus*.
  - (b) We have complete (known chilling and forcing) data for 329 angiosperm species and 107 gymnosperm species for the seed germination model.
  - (c) To prepare the egret data for our analysis we:
    - Separated warm and cold stratification duration into separate treatments
    - Substituted zero values when no cold stratification values are given
    - Transformed the percent germination to proportion germination
  - (d) To prepare the usda data for our analysis we:
    - When responses were reported as ranges we only used the maximum value in our analysis
    - When provided, day and nighttime germination temperatures were converted to weighted averages based on their reported photoperiods
2. We modeled budburst as a function of forcing, chilling, and photoperiod, accounting for phylogeny:

$$\hat{y}_i = \alpha_{sp[i]} + \beta_{force,sp[i]} \cdot forcing_i + \beta_{chill,sp[i]} \cdot chilling_i + \beta_{photo,sp[i]} \cdot photoperiod_i$$

where  $sp[i]$  is the species of observation  $i$ , and species-level intercepts and slopes ( $\alpha_{sp}$ ,  $\beta_{force,sp}$ ,  $\beta_{chill,sp}$ ,  $\beta_{photo,sp}$ ) are modeled according to species phylogenetic relationships.

3. We ran the model with forcing and chilling for seed germination considering phylogeny in the model. We took the values from the germination day temperature column as forcing and the values from the chilling duration column as chilling.
4. We modeled seed germination as a function of forcing and chilling, accounting for phylogeny, following the same structure as above (excluding the photoperiod term). The response variable was standardized to  $[0, 1]$  and modeled as an ordinal beta regression (with special handling for 0 and 1 observations).
5. After fitting both models, we extracted the species-level parameter estimates and compared the chilling and forcing effects on budburst versus seed germination for the 38 species overlapping between the two datasets which are all angiosperm species.